Raspberry Pi 5 Model B — cheat sheet

COMPETITION MASTER 01 — Raspberry Pi 5 Complete Smart Farm

Connections to remember

Controller pin	Goes to
3V3	MCP3008 · VREF
3V3 (3V3)	DHT22 · +
3V3 (3V3)	MCP3008 · VDD
5V	pH · VCC
5V	ACS712 pump current · VCC
5V	Relay ×4 · VCC
5V	Relay ×2 (dosing) · VCC
5V (5V)	MH-Z19B CO_2 · Vin
GND	DHT22 · –
GND	MH-Z19B CO_2 · GND
GND	MCP3008 · AGND
GND	MCP3008 · DGND
GND	pH · GND
GND	ACS712 pump current · GND
GND	Relay ×4 · GND
GND	Relay ×2 (dosing) · GND
GND	DOSING button · 2a
GND	TEST button · 2a
GPIO5 (GPIO5 dose A)	Relay ×2 (dosing) · IN1
GPIO6 (GPIO6 dose B)	Relay ×2 (dosing) · IN2
GPIO8 (CE0)	MCP3008 · CS/SHDN
GPIO9 (MISO)	MCP3008 · DOUT
GPIO10 (MOSI)	MCP3008 · DIN
GPIO11 (SCLK)	MCP3008 · CLK
GPIO14 (GPIO14 TX)	MH-Z19B CO_2 · RX
GPIO15 (GPIO15 RX)	MH-Z19B CO_2 · TX
GPIO16 (GPIO16)	DHT22 · out
GPIO17 (GPIO17 pump)	Relay ×4 · IN1
GPIO19 (GPIO19)	TEST button · 1a
GPIO22 (GPIO22 fan)	Relay ×4 · IN2
GPIO23 (GPIO23 alarm)	Relay ×4 · IN4
GPIO26 (GPIO26)	DOSING button · 1a
GPIO27 (GPIO27 light)	Relay ×4 · IN3

Settings

```
MAX_TEMP = 30  ·  FAN_OFF_TEMP = 27  ·  MIN_WATER = 20  ·  PUMP_SECONDS = 5  ·
IRRIGATION_HOURS = [7, 12, 17]  ·  LIGHT_ON_HOUR = 6  ·  LIGHT_OFF_HOUR = 20  ·  DOSE_A_ML =
10  ·  DOSE_B_ML = 5  ·  FLOW_ML_PER_SEC = 2  ·  PH_MIN = 5.5  ·  PH_MAX = 6.5  ·  EC_MIN
= 1.0  ·  EC_MAX = 2.5  ·  MIN_PUMP_AMPS = 0.2  ·  TEST_SECONDS = 2
```

Key lines of the program

main.py

```
dht = adafruit_dht.DHT22(board.D16)
co2_port = serial.Serial("/dev/serial0", 9600, timeout=1)
ph_adc = MCP3008(channel=0)
pump = OutputDevice(17)
dosing_button = Button(26)
    ph = round(ph_adc.voltage * 3.5, 2)
    pump_amps = round((current_adc.voltage - 2.5) / 0.185, 2)
    hour = time.localtime().tm_hour
app.add_url_rule("/status", "status", status)
    if dosing_button.is_pressed:
    if test_button.is_pressed:
```

The program in 6 steps

- Read the sensors (None/null when one fails).
- Buttons choose the mode: AUTO, DOSING, TEST.
- AUTO: fan 30/27 °C, light 06–20 h, irrigation at 7/12/17 h if water ≥ 20 %.
- DOSING: water check, B off, A for 5 s, A off, B for 2.5 s, B off.
- Alarms: SENSOR ERROR → LOW WATER → PUMP FAILURE → HIGH TEMPERATURE → OUT OF RANGE.
- Status every 5 s, wait 1 s, repeat.